Sources of Upper-Tail Human Capital: Evidence from Prussian Education Reforms: Online Appendices

Konstantin Bakharev, Roman Dimakov, Margarita Pavlova, Vladimir Shchukin

November 9, 2026

A. Regression results report: Births of Academics 1
A.1. Births: Counts of academics born in German states (All) 1
B. Regression results report: Deaths of Academics (unfinished) 4

Appendix A. Regression results report: Births of Academics

A.1. Births: Counts of academics born in German states (All)

TABLE A1. Effect of Prussian education reforms on academic births: baseline TWFE specifications

Dependent variable: Number of births of prominent academics (AcadEducHealth)				
	(1)	(2)	(3)	(4)
Prussia × Post-1790	-0.051	-0.060	0.031	0.031
	(0.036)	(0.037)	(0.041)	(0.041)
Estimator	Linear TWFE	Linear TWFE	Linear TWFE	Linear TWFE
City & Cohort FE	✓	✓	✓	✓
Population control	None	Time-varying log population	1750 log population and 1700–1750 growth	1750 log population and 1700–1750 growth
Population controls × Post	No	No	Yes	No
Population controls × Cohort	No	No	No	Yes
Weights	None	None	None	None
Sample	All cities	All cities	All cities	All cities
# Obs	7,722	7,650	7,592	7,592
R^2	0.359	0.360	0.393	0.400

Note. Standard errors clustered at the state level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A2. Effect of Prussian education reforms on academic births: PPML specifications

Dependent variable: Number of births of prominent academics (AcadEducHealth)			
	(1)	(2)	(3)
Prussia × Post-1790	0.253^{***}	0.210^{**}	0.199^{**}
	(0.093)	(0.088)	(0.094)
Estimator	PPML	PPML	PPML
City & Cohort FE	✓	✓	✓
Population control	None	Time-varying log population	1750 log population and 1700–1750 growth
Population controls × Post	No	No	Yes
Population controls × Cohort	No	No	No
Weights	None	None	None
Sample	All cities	All cities	All cities
# Obs	6,370	6,316	6,292
Pseudo R^2	0.276	0.276	0.276

Note. Standard errors clustered at the state level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A3. Effect of Prussian education reforms on academic births: population-weighted TWFE specifications

Dependent variable: Number of births of prominent academics (AcadEducHealth)			
	(1)	(2)	(3)
Prussia × Post-1790	1.047***	0.456***	0.210***
	(0.000)	(0.061)	(0.000)
Estimator	Linear TWFE	Linear TWFE	Linear TWFE
City & Cohort FE	City FE only	✓	City FE only
Population control	None	1750 log population and 1700–1750 growth	None
Population controls × Post	No	Yes	No
Population controls × Cohort	No	No	No
Weights	Analytic weights: population	Analytic weights: population	Analytic weights: log population
Sample	All cities	All cities	All cities
# Obs	7,650	7,592	7,650
R^2	0.622	0.716	0.367

Note. Standard errors clustered at the state level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A4. Effect of Prussian education reforms on academic births: alternative cohort aggregation

Dependent variable: Number of births of prominent academics (AcadEducHealth)				
	(1)	(2)	(3)	(4)
Prussia × Post-1790	-0.838	0.373	-0.102	0.063
	(0.717)	(0.819)	(0.073)	(0.084)
Estimator	Linear TWFE	Linear TWFE	Linear TWFE	Linear TWFE
City & Cohort FE	✓	✓	✓	✓
Population control	None	1750 log population and 1700–1750 growth	None	1750 log population and 1700–1750 growth
Population controls × Post	No	Yes	No	Yes
Population controls × Cohort	No	No	No	No
Weights	None	None	None	None
Sample	Two-period city panel	Two-period city panel	City by 10-year cohort	City by 10-year cohort
# Obs	594	584	3,861	3,796
R^2	0.693	0.777	0.485	0.531

Note. Columns (1)–(2) collapse the data into pre- and post-reform periods. Columns (3)–(4) use ten-year birth cohorts. Standard errors clustered at the state level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A5. Effect of Prussian education reforms on academic births: alternative city samples

Dependent variable: Number of births of prominent academics (AcadEducHealth)			
	(1)	(2)	(3)
Prussia × Post-1790	0.009	-0.034	-0.045
	(0.125)	(0.065)	(0.130)
Estimator	Linear TWFE	Linear TWFE	Linear TWFE
City & Cohort FE	✓	✓	✓
Population control	1750 log population and 1700–1750 growth	1750 log population and 1700–1750 growth	1750 log population and 1700–1750 growth
Population controls × Post	Yes	Yes	Yes
Population controls × Cohort	No	No	No
Weights	None	None	None
Sample	Cities above 15,000 baseline population	Cities with a pre-1790 academic birth	University cities
# Obs	780	3,796	884
R^2	0.556	0.401	0.317

Note. Standard errors clustered at the state level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

TABLE A6. Effect of Prussian education reforms on academic births: alternative state samples

Dependent variable: Number of births of prominent academics (AcadEducHealth)			
	(1)	(2)	(3)
Prussia × Post-1790	4.473^{***}	0.088^{**}	0.147^{**}
	(0.379)	(0.034)	(0.051)
Estimator	Linear TWFE	Linear TWFE	Linear TWFE
City & Cohort FE	✓	✓	✓
Population control	1750 log population and 1700–1750 growth	1750 log population and 1700–1750 growth	1750 log population and 1700–1750 growth
Population controls × Post	Yes	Yes	Yes
Population controls × Cohort	No	No	No
Weights	None	None	None
Sample	State capitals	Prussia and five largest states	Prussia and five selected small states
# Obs	702	6,318	4,186
R^2	0.638	0.420	0.470

Note. Standard errors clustered at the state level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Appendix B. Regression results report: Deaths of Academics (unfinished)

References